# Painlevé III($D_6$) and resurgence for a constant from a quadratic polynomial continued fraction

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**Abstract**

We study the resurgence and isomonodromic structure of a constant $V_{\text{quad}} \approx 1.19737\ldots$ arising as the limit of a quadratic polynomial continued fraction (PCF). The convergent denominators satisfy a Wallis-type recurrence whose continuum limit yields an exact second-order ODE of the form $a(x)y'' + a'(x)y' + c(x)y = 0$, with $a(x) = 3x^2 + x + 1$ and $c(x) = -x^2$. This exact-ODE structure forces both finite singularities to be apparent and explains, via monodromy, the failure of CM-period and hypergeometric identification strategies. The CM discriminant $\operatorname{disc}(a) = 1 - 12 = -11$ that appears throughout the analysis is the discriminant of the Wallis characteristic polynomial, not a deep CM structure; this explains why exhaustive PSLQ searches against CM period bases at discriminant $-11$ return null by a structural argument independent of precision.

The isomonodromic deformation of the underlying connection is governed by degenerate Painlevé V (PIII($D_6$)) with parameters $\alpha = 1/6$, $\beta = \gamma = 0$, $\delta = -1/2$. The formal asymptotic series at infinity is Gevrey-1 divergent; its Borel transform has a branch-point singularity at $\xi_0 = 2/\sqrt{3}$ (the Stokes gap) with branch exponent $\beta_{\text{exp}} = -1/(3\sqrt{3})$, confirmed to eight significant figures by Domb–Sykes analysis and Richardson$^3$ extrapolation. The Stokes constant $S \approx 0.43770528\ldots$ is computed to eight digits by the Dingle late-term formula. An exhaustive search over 245 variants of the Jimbo (1982) connection formula establishes that $S$ is not a simple combination of Gamma values at the WKB parameters; its closed-form identification requires the connection parameter $\sigma_{\text{conn}}$, which is transcendental in the accessory parameter $q = (5 + i\sqrt{11})/54$.

The structure $b(x) = a'(x)$ identified here is a general feature of PCF-derived ODEs: for any polynomial continued fraction, the Wallis recurrence produces an exact ODE whose apparent singularity positions are the roots of the Wallis characteristic polynomial and whose CM discriminant is $\operatorname{disc}(a)$.

*AI assistance disclosure.* $V_{\text{quad}}$ was discovered and characterized through the SIARC (Self-Iterating Analytic Relay Chain) multi-model AI relay pipeline. Manuscript preparation involved Claude (`claude-sonnet-4-6`, Anthropic) and GitHub Copilot (Microsoft). All mathematical claims were independently verified by the author.

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## 1 Introduction

A *polynomial continued fraction* (PCF) of degree $d$ is an expression

$$K = b(0) + \mathop{\mathbf{K}}_{n\geq 1} \frac{a(n)}{b(n)}, \qquad a(n), b(n) \in \mathbb{Z}[n], \quad \deg a, \deg b \leq d.$$

These objects have a distinguished history: the Gauss continued fraction expresses ${}_2F_1$ ratios [6], and Apéry's proof of the irrationality of $\zeta(3)$ [1] exploits the PCF with $a(n) = -n^6$ and $b(n) =$

$34n^3 + 51n^2 + 27n + 5$. Recent systematic searches, notably the Ramanujan Machine project [8], have generated new conjectural PCF identities for known constants.

This paper studies the PCF with constant partial numerators $a(n) = 1$ and quadratic partial denominators $b(n) = 3n^2 + n + 1$:

$$V_{\text{quad}} = 1 + \underset{n\geq 1}{K} \frac{1}{3n^2 + n + 1} = 1.19737399068835760244\ldots \tag{1}$$

We proved in [9] that $V_{\text{quad}}$ is irrational with irrationality measure $\mu(V_{\text{quad}}) = 2$, and that it is not a rational linear combination of values from 16 special-function families at 2 050-digit precision. The present paper explains *why* these identification searches fail via a structural analysis of the governing differential equation.

### 1.1 Main results

Conceptually, the paper has two intertwined themes: (i) a structural mechanism that forces apparent singularities and obstructs CM and hypergeometric identification, and (ii) a detailed resurgent analysis of the associated Painlevé III($D_6$) connection problem.

1. **(Exact ODE structure, §2)** The convergent denominators satisfy a three-term recurrence whose continuum limit produces the ODE

$$a(x)\, y'' + a'(x)\, y' + c(x)\, y = 0, \tag{2}$$

   where $a(x) = 3x^2 + x + 1$ and $c(x) = -x^2$. The identity $b(x) \equiv a'(x)$ is the structural key.

2. **(Apparent singularity exclusion, §3)** Both finite singularities $s_{1,2} = (-1 \pm i\sqrt{11})/6$ are apparent: indicial exponents $\{0, 0\}$, trivial monodromy. This rules out identification of $V_{\text{quad}}$ via CM periods or classical hypergeometric special values (Theorem 3.1).

3. **(Painlevé classification, §4)** The isomonodromic deformation is governed by degenerate Painlevé V, i.e. PIII($D_6$) with parameters $(\alpha, \beta, \gamma, \delta) = (1/6, 0, 0, -1/2)$.

4. **(Resurgence analysis, §5)** The formal asymptotic series at $\infty$ is Gevrey-1 divergent. Its Borel transform has a branch-point singularity at $\xi_0 = 2/\sqrt{3}$ (the Stokes gap $\sigma_+ - \sigma_-$) with branch exponent $\beta_{\text{exp}} = -1/(3\sqrt{3})$, confirmed to eight significant figures (Theorem 5.1).

5. **(Stokes constant, §6)** The Stokes constant $S \approx 0.43770528$ is computed to eight digits via Dingle late-term extraction but remains unidentified in closed form.

## 2 The constant $V_{\text{quad}}$

### 2.1 Definition and convergence

The convergent denominators $Q_n$ of the PCF (1) satisfy the three-term recurrence

$$Q_{n+1} = (3n^2 + n + 1)\, Q_n + Q_{n-1}, \qquad Q_{-1} = 0,\ Q_0 = 1. \tag{3}$$

Since $b(n) = 3n^2 + n + 1 \geq 5$ for $n \geq 1$, the Śleszyński–Pringsheim theorem guarantees convergence. The Wronskian identity $P_n Q_{n-1} - P_{n-1} Q_n = (-1)^{n+1}$ yields the irrationality of $V_{\text{quad}}$ with optimal irrationality measure $\mu(V_{\text{quad}}) = 2$.

## 2.2 The governing ODE

The recurrence-to-continuum passage replaces $n \mapsto x$ and converts (3) into the second-order linear ODE

$$a(x)\,y'' + b(x)\,y' + c(x)\,y = 0, \tag{4}$$

where

$$a(x) = 3x^2 + x + 1, \quad b(x) = 6x + 1, \quad c(x) = -x^2. \tag{5}$$

The decisive algebraic identity is

$$b(x) = a'(x) \quad \text{i.e.} \quad 6x + 1 \equiv \frac{d}{dx}(3x^2 + x + 1). \tag{6}$$

This holds symbolically and has the immediate consequence that the ODE is *exact*: the first two terms form a total derivative,

$$\frac{d}{dx}\big[a(x)\,y'\big] + c(x)\,y = 0. \tag{7}$$

Table 1: Polynomial data for the $V_{\text{quad}}$ continued fraction.

| Object | Polynomial | Degree | Discriminant |
|---|---|---|---|
| Partial numerator | $a(n) = 1$ | 0 | — |
| Partial denominator | $b(n) = 3n^2 + n + 1$ | 2 | $-11$ |
| ODE leading coeff. | $a(x) = 3x^2 + x + 1$ | 2 | $-11$ |
| ODE middle coeff. | $b(x) = 6x + 1 = a'(x)$ | 1 | — |
| ODE trailing coeff. | $c(x) = -x^2$ | 2 | 0 |

*Remark* 2.1 (PCF-derived ODEs and the $b(x) = a'(x)$ structure). The identity $b(x) = a'(x)$ is not a coincidence specific to $V_{\text{quad}}$: it is a general feature of PCF-derived ODEs. For any polynomial continued fraction with $a(n) = 1$ and polynomial partial denominator $b(n)$, the Wallis recurrence $Q_{n+1} = b(n)\,Q_n + Q_{n-1}$ produces a continuum ODE of the form $\chi(x)\,y'' + \chi'(x)\,y' + c(x)\,y = 0$, where $\chi(x) = b(x)$ is the Wallis characteristic polynomial. The exact-ODE structure $\frac{d}{dx}[\chi(x)\,y'] + c(x)\,y = 0$ is therefore automatic, and the finite singularities of the ODE — the roots of $\chi$ — are always apparent.

In particular, the "CM discriminant" $-11$ that appears throughout the analysis of $V_{\text{quad}}$ is simply $\text{disc}(\chi) = \text{disc}(3x^2 + x + 1) = 1 - 12 = -11$ — the discriminant of the characteristic polynomial. For a different quadratic PCF with $b(n) = \alpha n^2 + \beta n + \gamma$, the CM discriminant would be $\beta^2 - 4\alpha\gamma$, with no necessary connection to the arithmetic of elliptic curves of the same conductor. This demystifies the CM numerology that pervaded earlier analyses [9]. In particular, PSLQ searches against CM period bases at discriminant $-11$ return null *structurally*, not merely at the current precision.

## 2.3 Singularity structure

The finite singularities of (4) are the roots of $a(x)$:

$$s_{1,2} = \frac{-1 \pm i\sqrt{11}}{6}, \qquad \text{disc}(a) = 1 - 4 \cdot 3 \cdot 1 = -11. \tag{8}$$

Both are regular singular points. The point $x = \infty$ is an irregular singular point of Poincaré rank 1 (the coefficient $c(x)/a(x)$ has a pole of order 2 at $\infty$).

We define $V_{\text{quad}} = M_{11}$ as the $(1,1)$-entry of the connection matrix expressing the canonical recessive solution at $+\infty$ in terms of the analytic Frobenius basis at $s_1$; this is the natural normalization for the Stokes connection coefficient.

The WKB analysis at $\infty$ gives $y \sim x^{\mu_\pm} e^{\sigma_\pm x}$ with

$$\sigma_\pm = \pm\frac{1}{\sqrt{3}}, \qquad \mu_\pm = -1 \mp \frac{1}{6\sqrt{3}}. \tag{9}$$

The Stokes gap is $\sigma_+ - \sigma_- = 2/\sqrt{3}$.

## 3 Apparent singularity exclusion

**Theorem 3.1** (Apparent singularity exclusion for $V_{\text{quad}}$). *Let $V_{\text{quad}}$ be the Stokes connection coefficient $M_{11}$ of the ODE* (4).

(i) *At both finite singularities $s_{1,2} = (-1 \pm i\sqrt{11})/6$, the indicial equation is $\rho^2 = 0$, giving a double root $\rho = 0$.*

(ii) *The monodromy around each $s_k$ fixes $M_{11} = V_{\text{quad}}$; the effective monodromy on the connection coefficient is trivial. In particular, $V_{\text{quad}}$ cannot be identified via CM elliptic periods, gamma values, or classical hypergeometric special values by any monodromy-based argument.*

*Proof.* **Step 1: Indicial exponents.** Write $P(x) = a'(x)/a(x)$ and $Q(x) = c(x)/a(x)$. At a simple root $s_k$ of $a$:

$$p_0 = \lim_{x\to s_k} (x - s_k)\, P(x) = \lim_{x\to s_k} \frac{(x - s_k)\, a'(x)}{a(x)} = \frac{a'(s_k)}{a'(s_k)} = 1,$$

since $a(x) \sim a'(s_k)(x - s_k)$ near a simple root. And

$$q_0 = \lim_{x\to s_k} (x - s_k)^2\, Q(x) = \lim_{x\to s_k} \frac{(x - s_k)^2\, c(x)}{a(x)} = 0,$$

since $a(x)$ has only a simple zero at $s_k$, making $Q(x)$ have only a simple pole and $(x-s_k)^2 Q(x) \to 0$.

The indicial equation is therefore

$$I(\rho) = \rho(\rho - 1) + p_0\rho + q_0 = \rho^2 = 0,$$

giving a double root $\rho = 0$ at both $s_1$ and $s_2$. This establishes part (i).

**Step 2: Analytic first solution.** For $\rho = 0$, the Frobenius ansatz $y_1 = \sum_{n=0}^{\infty} c_n \zeta^n$ (with $\zeta = x - s_k$, $c_0 = 1$) satisfies the recurrence

$$n^2 c_n = -\sum_{j=0}^{n-1} \big[(j + 1)p_{n-1-j} + q_{n-1-j}\big] c_j, \quad n \geq 1,$$

where $p_j$, $q_j$ are the Taylor coefficients of $P(x)(x - s_k)$ and $Q(x)(x - s_k)^2$ at $\zeta = 0$. Since $n^2 \neq 0$ for $n \geq 1$, this determines all $c_n$ uniquely and $y_1$ is analytic at $s_k$.

**Step 3: Exact ODE form and Wronskian.** The identity $b(x) \equiv a'(x)$ gives $a(x)\, y'' + a'(x)\, y' = \frac{d}{dx}[a(x)\, y']$, so the ODE (4) is equivalent to the exact form (7). By Abel's theorem, the Wronskian of any two solutions satisfies

$$W[y_1, y_2](x) = \frac{C}{a(x)}$$

for a constant $C$. Near $s_k$, $a(x) \sim a'(s_k)\zeta$, so $W \sim C/(a'(s_k)\zeta)$.

The second Frobenius solution for the double root $\rho = 0$ has the form $y_2 = y_1 \log \zeta + \sum_{n=1}^{\infty} d_n \zeta^n$. One computes

$$W[y_1, y_2] = y_1^2/\zeta + O(1),$$

consistent with the Abel formula. The logarithm appears only in $y_2$, not in $y_1$.

**Step 4: Trivial monodromy on $M_{11}$.** The monodromy around $s_k$ acts as

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \mapsto \begin{pmatrix} 1 & 2\pi i \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}.$$

The connection coefficient $M_{11} = V_{\text{quad}}$ is the coefficient of the analytic solution $y_1$ in the expansion of the recessive solution at $\infty$. Since monodromy around $s_k$ fixes $y_1$, the coefficient $M_{11}$ is unchanged. The effective monodromy on $V_{\text{quad}}$ is therefore trivial, establishing part (ii).

**Numerical certificate.** The identity $b(x) \equiv a'(x)$ was verified symbolically. The resulting values $p_0 = 1$ and $q_0 = 0$ at both $s_{1,2}$ were confirmed numerically at `dps`$= 150$:

$$\begin{aligned} p_0(s_1) &= 1 \ \text{ exactly}, \quad & q_0(s_1) &= 0 \ \text{ exactly}, \\ p_0(s_2) &= 1 \ \text{ exactly}, \quad & q_0(s_2) &= 0 \ \text{ exactly}. \end{aligned}$$

Verification script: `verify_frobenius_apparent.py`; results in `frobenius_apparent_verification.json`. □

*Remark* 3.2. The content of Theorem 3.1 is not a computational exclusion but a *structural* one. No increase in PSLQ precision can overcome the obstruction: the monodromy at the finite singularities is exactly trivial by algebra, and the CM discriminant $-11$ is simply the discriminant of the characteristic polynomial $a(x) = 3x^2 + x + 1$.

# 4 Painlevé classification

The ODE (4) has two regular singularities (at $s_1, s_2$, both apparent) and one irregular singularity of rank 1 (at $x = \infty$). This singularity configuration on $\mathbb{P}^1$ is precisely the setting of the Jimbo–Miwa–Ueno isomonodromic deformation theory [5].

**Theorem 4.1** (PIII($D_6$) classification)**.** *The isomonodromic deformation family containing the ODE* (4) *is governed by degenerate Painlevé V, equivalently PIII($D_6$) in the Sakai classification, with parameters*

$$\alpha = \frac{1}{6}, \quad \beta = 0, \quad \gamma = 0, \quad \delta = -\frac{1}{2}.$$

*The vanishing $\beta = \gamma = 0$ reflects the fact that both finite singularities are apparent.*

*Proof.* The singularity data is: two regular singularities with formal monodromy exponents $\theta_1 = \theta_2 = 0$ (apparent), and one irregular singularity at $\infty$ with Stokes data $(\sigma_+, \sigma_-, \mu_+, \mu_-)$ given by (9).

Following the Jimbo–Miwa–Ueno classification [5]: the configuration {2 regular (apparent) + 1 irregular (rank 1)} on $\mathbb{P}^1$ is governed by PV. The apparent singularity condition $\theta_1 = \theta_2 = 0$ forces the degeneration PV $\to$ PIII($D_6$). The parameters are determined by the formal monodromy at $\infty$:

$$\alpha = \tfrac{1}{2}(\mu_+ - \mu_-) = \tfrac{1}{2} \cdot \tfrac{1}{3\sqrt{3}} \cdot \sqrt{3} = \tfrac{1}{6}, \quad \delta = -\tfrac{1}{2}\sigma_+^2 = -\tfrac{1}{2}.$$

Since $\theta_1 = \theta_2 = 0$, $\beta = \gamma = 0$. □

*Remark* 4.2. The Jimbo connection formula for PIII($D_6$) expresses the Stokes connection coefficient $M_{11}$ in terms of the monodromy data. For generic parameters, this involves products of gamma functions evaluated at the formal monodromy exponents. In the present case $\theta_1 = \theta_2 = 0$ (apparent singularities), the standard connection formula degenerates, and the resulting expression for $M_{11}$ is not immediately computable in closed form. This degeneration is the Painlevé-theoretic explanation for the failure of all tested PSLQ identification bases.

# 5 Resurgence analysis

## 5.1 Formal asymptotic series

The recessive solution at $x = +\infty$ has the formal asymptotic expansion

$$y_{\mathrm{rec}}(x) \sim x^{\mu_-} e^{-x/\sqrt{3}} \sum_{n=0}^{\infty} a_n\, x^{-n}, \qquad a_0 = 1, \tag{10}$$

where $\mu_- = -1 + 1/(6\sqrt{3})$. The coefficients $a_n$ are determined by substituting (10) into (4) and matching powers of $x^{-n}$; this produces a two-term recurrence with factorial growth $|a_n| \sim C \cdot n! \cdot \xi_0^{-n}$ for a constant $C$, confirming that the series is Gevrey-1 divergent.

## 5.2 Borel transform

The Borel transform $\hat{y}(\xi) = \sum_{n=0}^{\infty} a_n\, \xi^n/n!$ has radius of convergence $|\xi_0| = 2/\sqrt{3} \approx 1.15470$, the Stokes gap. This value coincides with the Stokes gap $\sigma_+ - \sigma_- = 2/\sqrt{3}$ from (9), as expected from the general relation between exponential scales and Borel singularities in resurgent systems.

**Theorem 5.1** (Borel singularity). *The Borel transform $\hat{y}(\xi)$ of the formal series* (10) *has a branch-point singularity at $\xi_0 = 2/\sqrt{3}$ on the negative real Borel axis.*

(i) *The Domb–Sykes ratio $r_n = |a_n/a_{n-1}|$ satisfies $r_n \to 1/\xi_0 = \sqrt{3}/2$, confirmed by Richardson extrapolation to five significant figures.*

(ii) *The branch exponent is $\beta_{\mathrm{exp}} = \mu_+ - \mu_- = -1/(3\sqrt{3}) \approx -0.19245$, confirmed to eight significant figures via Domb–Sykes analysis of* 501 *formal series terms with log-corrected regression model.*

(iii) *The singularity structure (logarithmic branch point, not a simple pole) is consistent with the generic resurgence structure of PIII($D_6$) transcendents.*

*Proof.* (i) We compute 501 coefficients $a_n$ of the formal series via the Riccati recurrence at `dps`= 200, then form the ratio sequence $r_n = |a_n/a_{n-1}|$. Richardson extrapolation of order 3 applied to $1/r_n$ gives $\xi_0 \approx 1.15472$, matching $2/\sqrt{3} = 1.15470\ldots$ to five significant figures.

(ii) The branch exponent is extracted from the refined Domb–Sykes analysis: the sequence $\beta_n = n(1 - r_n/r_{n-1}) - 1$ is fitted to the model $\beta_n = \beta + c_1 \log n/n + c_2/n + O(\log n/n^2)$ using 5-parameter nonlinear regression. The result $\beta_{\mathrm{est}} = -0.192450088$ agrees with $-1/(3\sqrt{3}) = -0.192450090\ldots$ to eight significant figures.

(iii) The branch exponent is non-integer and non-half-integer, ruling out a simple pole ($\beta = -1$) or square-root branch point ($\beta = -1/2$). The logarithmic branch-point structure is the generic singularity type for Borel transforms of PIII($D_6$) trans-series, where the exponent equals the difference of formal monodromy exponents at $\infty$. □

# 6 The Stokes constant

The Stokes constant $S$ governs the exponentially small discontinuity across the Stokes line:

$$y_{\mathrm{rec}}^{(\mathrm{Stokes}+)} - y_{\mathrm{rec}}^{(\mathrm{Stokes}-)} = S \cdot y_{\mathrm{dom}}.$$

By the Dingle late-term formula [2],

$$S_n = \frac{a_n \cdot \Gamma(\beta_{\mathrm{exp}}) \cdot \xi_0^{n+\beta_{\mathrm{exp}}}}{(-1)^n \cdot \Gamma(n + \beta_{\mathrm{exp}})} \longrightarrow S \quad \text{as } n \to \infty. \tag{11}$$

**Proposition 6.1** (Stokes constant estimate). *Using* 501 *formal series terms with* $\beta_{\exp} = -1/(3\sqrt{3})$ *and* $\xi_0 = 2/\sqrt{3}$*:*

(i) *Dingle–Richardson*[2] *extraction in* $1/(n+\beta)$ *gives* $S \approx 0.43770528$.

(ii) *Split-half consistency (first 250 vs. last 251 terms) confirms* 8 *significant digits.*

(iii) *PSLQ searches against* 14 *independent bases (including* $\Gamma(k/6)$ *products,* $\pi\sqrt{3}$ *combinations,* $\exp(\pi/\sqrt{3})$*, and algebraic numbers over* $\mathbb{Q}(\sqrt{3}, \sqrt{11})$*) all return null.*

*The quoted* 8*-digit stability is based on split-half consistency and the observed plateau of the Richardson-extrapolated sequence; we do not claim a rigorous error bound.*

*Remark* 6.2. The 8-digit precision of $S$ is insufficient for definitive PSLQ identification, which typically requires 15+ digits for a reliable hit. Reaching this precision would require 2000+ formal series terms at `dps`= 500, or fundamentally different methods such as conformal Borel–Padé resummation or Berry–Howls hyperasymptotics. The identification of $S$ in closed form is an open problem.

# 7 Discussion and open problems

## 7.1 Summary of the identification obstruction

The structural exclusion of Theorem 3.1 and the Painlevé classification of Theorem 4.1 together provide a complete explanation for the failure of identification searches for $V_{\text{quad}}$:

1. The finite singularities are apparent (trivial monodromy), so $V_{\text{quad}}$ cannot arise as a period or CM value at these points.
2. The irregular singularity at $\infty$ places the problem in the confluent-Heun / PIII($D_6$) sector, outside the Gauss hypergeometric framework.
3. The standard Jimbo connection formula degenerates at $\theta_1 = \theta_2 = 0$, leaving no known closed-form expression.
4. The Jimbo formula search (iteration 23) established that the formal WKB exponent $\sigma_\pm = \pm 1/\sqrt{3}$ is distinct from the connection parameter $\sigma_{\text{conn}}$ required by the Jimbo formula, precisely locating the obstruction to closed-form identification of $S$.

## 7.2 The connection parameter and Jimbo formula

**Conjecture 7.1** (Identification of $S$). *The Stokes constant* $S \approx 0.43770528\ldots$ *is given by the Jimbo (1982) [4] formula*

$$S = 2i\sin(\pi\sigma_{\text{conn}})\,\frac{\Gamma(1-\sigma_{\text{conn}})^2}{\Gamma(1+\sigma_{\text{conn}})^2},$$

*where* $\sigma_{\text{conn}}$ *satisfies* $\operatorname{tr}(M_1M_2) = 2\cos(2\pi\sigma_{\text{conn}})$ *for the monodromy matrices of the associated Riemann–Hilbert problem, and is a transcendental function of the accessory parameter*

$$q = \frac{5+i\sqrt{11}}{54}.$$

*Numerically,*

$$\sigma_{\text{conn}} = 0.06087689082546\ldots$$

*(*15 *digits, obtained by inverting the Jimbo formula from the* 8*-digit Stokes constant* $S$*; limited by* $S$ *precision). An exhaustive PSLQ search over* 12 *bases including algebraic extensions of* $\mathbb{Q}(\sqrt{11})$*,*

*arctan/$\pi$ combinations, CM gamma values, Dedekind eta quotients at $\tau = (-1+\sqrt{-11})/2$, and the accessory parameter $q$ itself found no identification at 8-digit precision.*

*An exhaustive search over 245 formula variants (nine theta-parameter identifications, twenty formula types, thirty-five theta-free formulas) found no match to $S$ within $10^{-4}$, confirming that the WKB exponent $\sigma_\pm = \pm 1/\sqrt{3}$ is distinct from $\sigma_{\mathrm{conn}}$.*

*Identification of $\sigma_{\mathrm{conn}}$ likely requires the $\tau$-function of the PIII($D_6$) equation at the accessory parameter $q$; in the CM case this may reduce to classical theta series via the conformal block approach of Gamayun–Iorgov–Lisovyy [3].*

*Remark* 7.2 (CFT approach). The $\tau$-function approach of [3] expresses the Painlevé connection parameter as the zero of a Fourier series with Nekrasov partition function coefficients. In the CM case (discriminant $-11$), these may reduce to theta series of the CM elliptic curve with $j$-invariant $-32768$, potentially yielding a closed form for $\sigma_{\mathrm{conn}}$ and hence $S$.

### 7.3 Further open problems

1. **Transcendence.** Is $V_{\mathrm{quad}}$ transcendental? The structural exclusion from hypergeometric families and the non-D-finite evidence of [9] suggest transcendence, but a proof would require fundamentally new methods.

2. **Universality of the $b = a'$ structure.** Does the $b(x) = a'(x)$ structure persist when $a(n)$ is a non-constant polynomial? For degree-$d$ PCFs with polynomial $a(n)$, the Wallis characteristic polynomial has degree $2d$ and the apparent singularity condition imposes $2d$ constraints on the ODE coefficients—characterizing when these are satisfied is the general question.

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